

The invisible arms race: digital trends in illicit goods trafficking and AI-enabled responses

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Abstract—Recent trends in the *modus operandi* of technologically-aware criminal groups engaged in illicit goods trafficking (e.g., firearms, drugs, cultural artifacts, etc.) have given rise to significant security challenges. The use of cryptocurrency-based payments, 3D printing, social media and/or the Dark Web by organized crime leads to transactions beyond the reach of authorities, thus opening up new business opportunities to criminal actors at the expense of the greater societal good and the rule of law. As a result, a lot of scientific effort has been expended on handling these challenges, with Artificial Intelligence (AI) at the forefront of this quest; mostly machine learning and data mining methods that can automate large-scale information analysis. Deep Neural Networks (DNNs) and graph analytics have been employed for automatically monitoring and analyzing the digital activities of large criminal networks in a data-driven manner. However, such practices unavoidably give rise to ethical and legal issues, which need to be properly considered and mitigated. This paper is the first one to systematically explore all of the above aspects jointly and from a combined perspective, without focusing on a particular angle or

type of illicit goods trafficking. It emphasizes how advances in AI both allow the authorities to unravel technologically-aware trafficking networks and provide countermeasures against any potential violations of citizen rights in the name of security.

Index Terms—3D printing, artificial intelligence, AI ethics, cryptocurrency, Dark Web, Deep Neural Networks, graph analytics, security, trafficking, trustworthy AI.

I. INTRODUCTION

RECENT developments in digital technologies have greatly facilitated trans-border trafficking of illicit goods. During the past decade, it has been made clear that the exchange of drugs, firearms and other contraband is made significantly easier by social media, Dark Web marketplaces, cryptocurrency transactions, 3D printers and similarly sophisticated tools. As a result, the unlawful activities of organized crime networks and terrorist/extremist groups can stay beyond the reach of the law for longer time periods with fewer difficulties.

This situation has increased the stakes for Law Enforcement Agencies (LEAs) and/or public authorities, giving rise to a progressively worsening security environment across the globe. The consequences can already be felt even in politically stable liberal regimes with an advanced economy: the erosion of the rule of law, the wider spread of corruption, the hindering of economic growth, the undermining of people's trust to legitimate authorities and the abrasion of their fundamental rights to safety and security are potential side-effects of these developments [1].

The harmful societal and economic impact of illicit goods trafficking, such as firearms or drugs, is obviously enormous. For example, in 2019 alone, more than 250 thousand people died as a result of firearms worldwide, where nearly 71% of gun deaths were homicides, about 21% were suicides and 8% were unintentional firearms-related accidents¹. On the other hand, in the European Union alone, over 8,300 deaths involving one or more illicit drugs were reported in 2018², while roughly EUR 3.3 billion are spent on an annual basis for hospital-based drug treatment. More generally, illicit trafficking activities are intricately linked to various forms of violent organized crime and provide funding or manpower for multiple criminal/terrorist

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¹<https://worldpopulationreview.com/country-rankings/gun-deaths-by-country>

²https://www.emcdda.europa.eu/system/files/publications/13762/TD0221591E_NN.pdf

groups³.

Profit- or ideology-driven networks engaging in illicit trafficking are typically interconnected, geographically dispersed, fluid, secretive and flexible in their structure. Thus, the very nature of on-line transactions, coupled with the inherent anonymity offered by cryptocurrency payments and by onion routing-based networking software – like Tor, which is commonly employed for accessing the Dark Web – renders such options highly attractive to them. As a result, novel criminal *modi operandi* have emerged during the past decade. In certain cases traffickers can operate not only through the Dark Web, but also through regular Surface Web e-commerce sites; this is true, for instance, in the case of legally purchased dual-use and (pre-)pre-cursor chemical substances that can be abused by the byer for producing illegal drugs [2]. A more exotic practice that is currently becoming more common is to digitally distribute blueprints of desired contraband (e.g., guns), which are then 3D printed using appropriate equipment. In such cases, only specific parts which cannot be reproduced by 3D printing (e.g., metal weapon barrels) need to be physically smuggled through shipping containers, or even via parcels sent through legitimate postal/courier services.

Advanced Artificial Intelligence (AI) algorithms for large-scale information processing hold the potential to facilitate the detection, analysis, monitoring, reconstruction and unraveling of such technologically-aware trafficking groups, by identifying and correlating their activities and the involved actors. Therefore, AI can serve as a valuable tool for enhancing LEA productivity, response times and ability to counteract the appropriation of high-tech by organized crime. This mostly holds true for modern machine learning and data mining methods, such as Deep Neural Networks (DNNs) and graph analytics, which are able to sift through massive amount of information in order to automatically detect actionable patterns, insights and trends in minimal time.

However, advanced AI-enabled technology operating on massively collected, large-scale real-world data at the hands of state authorities gives rise to significant ethical and legal issues: the citizens' right to privacy may be placed at risk, inherent bias within machine learning models may reproduce systemic inequalities, while constitutional guarantees may be jeopardized. A potential technical solution to these concerns arises from the fledgling scientific subfield of Trustworthy AI, which promises to mitigate similar dangers by imparting AI algorithms with inherent robustness against them.

Overall, this paper explores the above issues systematically and synergistically, emphasizing how advances in AI *both* allow the authorities to unravel technologically-aware trafficking networks *and* provide countermeasures against any potential violations of citizen rights in the name of security. To the best of the authors' knowledge, no previous recent article either investigates this domain from such a combined perspective, or treats all illicit goods trafficking activities at once. Existing papers tend to focus only on a specific type of illicit

trafficking and its particularities [3] [4], do not always engage with modern digital trends employed by traffickers and deal with either exclusively technical [5] [6], exclusively criminological [7] or exclusively ethical/legal issues [8] [9].

The remainder of this article is structured in the following manner. Section II presents and discusses recent criminal *modi operandi* that have emerged in illicit goods trafficking, thanks to modern digital technologies. Section III briefly overviews state-of-the-art AI methods that can facilitate the detection, analysis, monitoring, reconstruction and unraveling of technologically-aware trafficking networks. This discussion is *not* deeply technical and emphasizes how AI algorithms for large-scale information processing can be employed to assist LEAs. Section IV reviews legal and ethical concerns that arise from the extensive employment of AI by LEAs, along with potential mitigation countermeasures offered by Trustworthy AI methods. Finally, Section V draws conclusions from the preceding discussion.

II. EMERGING CRIMINAL MODI OPERANDI EXPLOITING DIGITAL TECHNOLOGIES

The advent of modern digital technologies has revolutionised the way in which certain “traditional” crimes are perpetrated. For example, the World Wide Web, with its three main layers (Surface Web, Deep Web and Dark Web) has given rise to entirely new forms of criminal activity (e.g., phishing, the use of malware or ransomware, etc.), but it has also altered the *modi operandi* used to commit crimes. It has multiplied the avenues for financial fraud, the access to vulnerable children by child-sex offenders, and the possibilities for illicitly trafficking goods such as drugs and firearms [10].

The Dark Web is a part of the Deep Web accessible only through specific browsers, usually Tor, which ensures anonymity to the users by essentially hiding their IP address from surveillance and traffic analysis while navigating. The anonymity provided by the Dark Web makes it attractive to individuals interested in carrying out criminal activities with a low risk of being tracked and identified, including the sale and purchase of illegal goods [11] [12] [13].

Within the Dark Web, two types of marketplaces are employed to traffic illicit goods: vendor shops and cryptomarkets. Also known as “single-vendor markets”, vendor shops are on-line stores administered by a single seller, who directly manages the transactions with customers, without third-party services. The direct connection between seller and buyer decreases commissions' costs and risks associated with third-party services [14] [15]. According to Europol [16], in recent years there has been a rise in single-vendor shops in the Dark Web, which frequently rely also on encrypted communication platforms, such as Wickr and Telegram, to manage sales. These findings highlight a tendency in criminal activities towards more decentralized and secure channels that ensure higher trust among involved parties.

On the other hand, similarly to legal e-commerce platforms on the Surface Web, cryptomarkets offer the following possibilities: a) for multiple vendors to display their goods

³https://www.europol.europa.eu/cms/sites/default/files/documents/business_fundamentals_how_illegal_drugs_sustain_organised_crime_in_europe.pdf

available for sale by posting images and descriptions of the goods, and b) for buyers to search for a specific good. Cryptomarkets provide more visibility in comparison to single-vendor shops, so access to a broader client base for sellers, and third-party services to make transactions safer for the buyer. An example is the escrow, which ensures that the cryptocurrency payment is held by the market until the buyer confirms reception of the goods, or a designated waiting period has elapsed, at which point the transaction is made accessible to the vendor. Additionally, this way the seller preserves his/her anonymity and minimizes the chance of the illicit transaction being traced back to him/her [14] [11] [18] [19] [20] [13]. The quality and the assurance of delivery are then assessed by buyers, who are thus able to impact vendors' reputation [21].

Dark Web markets, and more generally the Internet, have allowed for a relevant innovation in illicit goods trafficking by decreasing the effort required to conduct it [22] [23]. Illicit goods, such as drugs and firearms, can be ordered and paid just by clicking a button, while being received in a few days across the world. Buyers and sellers do not need to meet at a physical location; they can conclude more illicit transactions via the Internet, in a quicker and safer manner and without sharing their identities or contact information [20] [13]. The on-line availability of illicit goods allows an easier matching of demand and supply in the case of inexperienced individuals with no direct or indirect criminal connections, which is instead a fundamental feature of off-line sales [24] [25].

Goods sold on-line are delivered mainly through postal and courier services, which allow vendors to easily and quickly send goods anywhere in the world. Shipments in this case are quite small (multiple small shipments can be sent at a time) and the addresses of the recipients do not correspond to personal addresses of the buyers [26]. Other methods used include concealing goods inside legitimate items and making the parcel appear professional [26] [27]. Indeed, on-line marketplace forums are used by vendors to discuss about shipping options for avoiding law enforcement detection, based on criteria for profiling suspect packages that are regularly published by governments [14]. Nonetheless, in recent years, vendors and buyers have started avoiding traditional courier shipments when possible, relying on the so-called "dead drop" method. It involves paid intermediaries who first conceal the prepackaged illicit shipment in discreet locations, and then share with the vendor the coordinates through a short video for each "dropped" deal. The vendor then sends the geo-coordinates to the client, who can eventually pick up the goods [18] [14].

It is estimated that approximately 62% of the illicit goods sold in Dark Web markets are drugs and drug-related chemicals [28]. The on-line trade of other contraband, such as firearms, remains limited in volume and in value [29]. In fact, firearms and explosives account for only about 1% of illicit goods smuggled on the Dark Web [28]. Over the years, the sale of these goods has shifted from Dark Web marketplaces to Surface Web forums and secure platforms, such as Telegram and Wickr, following prohibitions on firearms sales

in several marketplaces [29] [16]. Within the European Union, relevant discussions on on-line forums and social media, such as Twitter, YouTube, Reddit, and Discord, mainly concern guidelines and instructions for manufacturing firearms at home, especially with the use of 3D printers by "enthusiasts"; this is an emerging modus operandi [30] [14]. Such digital products, also available in Dark Web markets as blueprint files for sale [16], are associated with even lower risks due to the lack of any physical exchange [14].

Overall, despite their currently smaller size compared to the traditional off-line market, Surface and Dark Web may become a more important marketplace in the near future for the trafficking of illicit goods other than drugs. The use of on-line channels increased a lot, for example, due to the restrictions caused by the Covid-19 pandemic, which led to a re-orientation of logistical chains. However, scarce information is available on the long-term impact of the pandemic on trafficking patterns [31].

Illicit drugs trafficking is clearly the domain most affected by digital technologies, with the variety of on-line exchange avenues being greater than in the case of other types of contraband. On the Dark Web, drug sales can occur within a marketplace, within a decentralized network or between individuals [32]. The most traditional recreational drugs such as cannabis, MDMA and LSD, as well as certain prescribed medicines, are the most popular in marketplaces [33]. Sales of new psychoactive substances and legal pharmaceutical products are traded mainly on the Surface Web [32]. Overall, four primary avenues have been identified: a) e-shops selling new psychoactive substances as research chemicals, mostly under their chemical names; b) e-shops selling products under brand names; c) classified ads, often located within public Web sites; and d) a Deep Web avenue [34]. These routes suggest a growing hybridization between the commercial and research chemical areas and the presence of a "grey market", which includes Web sites having both a Surface Web presence and a hidden Deep Web element [32].

The use of cryptomarkets also increased during the Covid-19 pandemic [35]; the physical lockdown measures pushed many people to order drugs on-line. However, the restrictions imposed by many governments may have negatively impacted the function of these markets as well, especially concerning international sales, since drug dealers mainly hide illicit drug loads within legitimate international shipments [17]. In fact, the number of unsuccessful transactions on cryptomarkets increased during the pandemic due to delivery failures, related to the international/inter-continental nature of the transactions and the severity of the crisis in the vendor's country [36]. Despite this, the opportunities offered by the Internet continue to foster the proliferation of the on-line drug markets. The closure of major Dark Web-hosted platforms seems to have a minimal long-term impact, since vendors can easily migrate to other platforms [37].

Fig. 1 summarizes the current state of the criminal modus operandi in illicit goods trafficking, emphasizing contemporary digital trends.

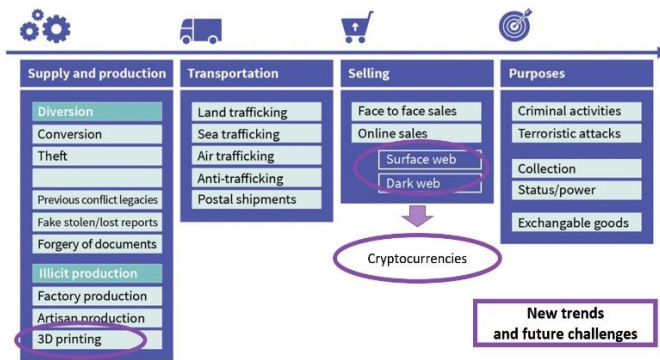


Fig. 1. A visual summary of the current criminal modus operandi in illicit goods trafficking.

III. ARTIFICIAL INTELLIGENCE FOR FIGHTING ILLICIT GOODS TRAFFICKING

State-of-the-art AI methods for large-scale information processing can be successfully leveraged for detecting, analyzing, monitoring, reconstructing and unraveling technologically-aware trafficking networks, which follow the trends presented in Section II. The methods most relevant to this application domain can be broadly classified into two categories:

- Deep Neural Networks (DNNs),
- Graph analytics.

Graph Neural Networks (GNNs) essentially form a hybrid approach, being neural networks that allow knowledge mining on graphs.

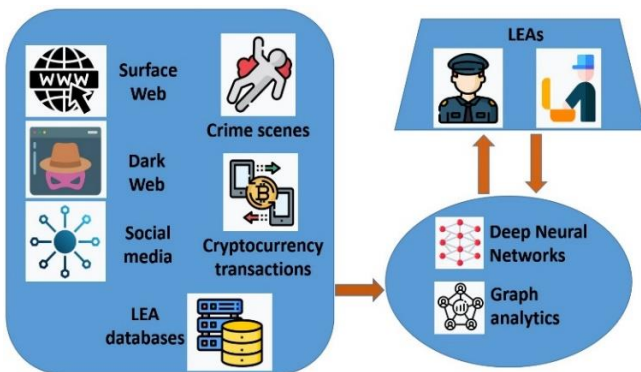


Fig. 2. High-level functional diagram of AI deployment for fighting illicit goods trafficking.

The following two Subsections present a high-level overview of both technologies, while focusing on applications relevant to fighting illicit goods trafficking. Fig. 2 depicts a functional diagram of how the presented technologies can be deployed in such a context.

A. Deep Neural Networks

DNNs are complex machine learning models that are trained to learn a desired input-output mapping. They are feed-forward, layered neural networks with a high number of learned parameters; typically in the range of millions or

billions. Training is usually performed on a large annotated dataset of known input vectors, using variants of the error back-propagation and stochastic gradient descent algorithms, with the parameters “freezing” when training finishes. From then on, the trained DNN model is ready to analyze novel test input vectors at an inference stage, to make useful predictions about them. The majority of useful downstream AI tasks are either classification tasks or regression tasks, with certain ones (e.g., object detection on images) combining aspects of both.

DNNs have a definite edge over traditional machine learning methods, thanks to their ability to automatically learn features from input data, which eliminates the need for manual feature extraction. Different neural architectures have been proposed over the years, which typically differ with regard to how their layers operate. Thus, Convolutional Neural Networks (CNNs) are mostly suitable for image analysis, since they receive their input in the form of tensors instead of vectors. Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks are mostly employed for analyzing time series and other types of sequences, thanks to their limited internal memory. Currently, Transformer architectures relying on the so-called “self-attention” mechanism empower the state-of-the-art algorithms in most AI tasks. Their strength lies in their ability to efficiently capture long-range dependencies and contextual information within sequential data of any nature.

DNNs have proven extremely effective in perception and language analysis tasks, particularly if there is a high volume of data to be analyzed. Naturally, their usual role with regard to fighting illicit goods trafficking falls under this area. The majority of relevant work exploits DNNs for object detection in image/video data, object recognition in images and for Natural Language Processing (NLP). The relevant data are typically massive in volume and gathered from the following sources:

- **Surface Web.** Web sites/fora, e-commerce sites. Typically, the goal of analyzing data from the Surface Web is to detect/monitor legitimate e-shops selling products that may facilitate illicit goods trafficking (e.g., licit drug precursors), identify discussions among users related to trafficking networks, or even discover blueprints for 3D printing of illicit goods (e.g., guns).
- **Dark Web.** Dark Web marketplaces. Typically, the goal of analyzing data from the Dark Web (e.g., images, text, etc.) is to directly identify illicit marketplaces where trafficking networks may thrive. Of course, given that the Dark Web is by definition concealed, it is not always easy to access relevant data and collect them for AI-enabled analysis, but dedicated Dark Web crawlers facilitate the process [38]. In a complementary fashion, Darknet network traffic data can also be collected and analyzed, even allowing real-time monitoring.
- **Social media.** Social media platforms, messaging apps. These are typically analyzed in order to identify discussions about and/or covert marketing/advertising of illicit goods, while also flagging users with an active interest in them. Text and images are the usual data

modalities extracted from such sources.

- **Crime scenes or entry points.** These are typically images/videos, along with relevant metadata, that are physically captured at crime scenes (e.g., at drug or gun shipment seizures) or on potential entry points for traffickers (e.g., customs offices, ship ports, airports, etc.). In the case of entry points, relevant data may have been captured en masse for prevention purposes (e.g., CCTV monitoring, X-ray scanning of all mailed parcels, millimeter-wave imaging of passengers, etc.). In rarer cases, data modalities may extend beyond the “visual + metadata” domain; for instance, chemical signatures derived from drug screening kits, results from vapor-based inspection of maritime containers, toxicology reports related to drug overdoses, etc.
- **LEA databases.** These are typically structured or unstructured data maintained by LEAs, both historical and current. They may contain almost any type of modality that can be analyzed by DNNs (text, audio, visual, etc.), potentially along with relevant metadata. The original data sources may also greatly vary, ranging from targeted wiretaps to massively crawled social media posts.

The following paragraphs briefly overview the most important aspects of the relevant state-of-the-art, organized per data modality.

A1. Visual data analysis

Regarding the fight against illicit goods trafficking, the typical visual data of interest are RGB images, RGB videos and X-ray scans or millimeter-wave images. DNNs are naturally at the forefront of relevant automated analysis.

Regular RGB image analysis has been extended to the case of illicit goods trafficking, with the vast majority of literature focusing on object recognition or object detection [39]. Since there is a considerable domain and task overlap with public safety applications (e.g., real-time monitoring of public spaces to prevent terrorist attacks, surveillance of general criminal activities, etc.), the volume of related research is large. However, the dominant trend is to simply adapt popular DNN solutions for generic image analysis, using image datasets depicting illicit goods. The application-specific challenges are mostly three: a) scarcity of very large, annotated relevant datasets, b) small on-image size of many objects-of-interest, and c) large variation in viewpoints, background clutter and levels of visual occlusion between different images depicting the same type of illicit good [40]. Additionally, unlike public safety monitoring, there is rarely an actual need for real-time processing. Most recent work of this kind employs CNNs for whole-image classification [41] [42] or for object detection purposes [43] [44] [45]. Well-known relevant neural architectures are typically utilized and properly adapted, such as VGG-16, YOLOv3-5, SSD, Faster R-CNN, with the choice mainly dictated by the desired inference speed-accuracy trade-off and the particularities of the specific dataset employed. In general, less complex architectures are faster and less prone to overfitting when trained with comparatively small datasets, but more complex DNNs may achieve greater test-stage

accuracy when trained on sufficiently large datasets.

Very few recent approaches deviate from the paradigm described above; this also holds true for RGB video analysis (mainly CCTV footage), where video frames are simply extracted and analyzed as separate images [46]. At most, each video frame may first be preprocessed to highlight any human bodies visible in the image, using a separate, pretrained DNN for person detection [47] or for human body pose estimation [48] [49]. A notable example that deviates from the norm is the method in [50], which exploits an ensemble of multiple, simple CNNs, instead of one monolithic DNN, for weapon detection in images. Each model detects a specific firearm component/part (e.g., barrel) and their outputs are aggregated to obtain a single final prediction.

DNNs also dominate modern scan image analysis for the X-ray and millimeter-wave modalities. The typical goal is to automatically detect illicit goods, such as drugs or weapons, in passengers, luggage or mailed parcels; an example is shown Fig. 3. The dominant trends are similar to those of the RGB image analysis (e.g., [51] [52]), but obviously different training datasets are utilized. However, special neural modules are commonly employed as part of the overall DNN [53] [54] [55] [56], so that accuracy is improved in the face of typical issues such as occlusions, cluttered background or class imbalance.

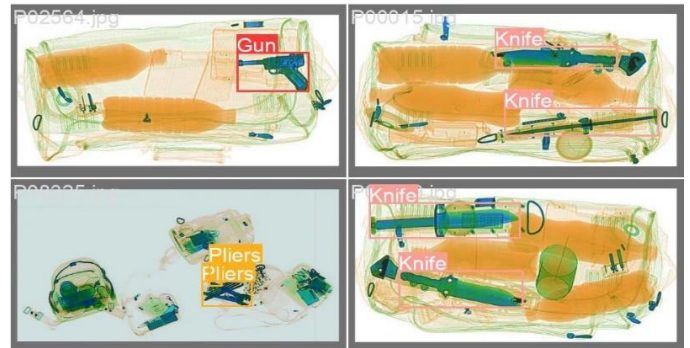


Fig. 3. An example of illicit item detection on an X-ray scan of a passenger luggage, using DNNs [39].

Finally, a few more idiosyncratic attempts have been made towards exploiting visual analysis DNNs to more explicitly identify on-line criminal actors. For instance, the method in [57] exploits CNNs for illicit vendor re-identification/fingerprinting in the Dark Web, through recognizing the style of their posted photographs.

A2. Language/text analysis

NLP algorithms are mainly utilized to identify discussions, user/customer reviews or advertisements concerning illicit goods, or facilitators for their production (e.g., drug precursors, 3D printing blueprints of firearms). The data sources can be any type of on-line content, text from LEA databases or transcriptions of audio stored in LEA databases (e.g., from wiretaps). The vast majority of relevant literature concerns classification of short texts as being about specific

types of illegal activities (e.g., [58] for detecting drug dealers).

DNNs dominate the state-of-the-art, as in the majority of NLP tasks. The current trend is to use Transformer architectures that have been pretrained as language models in a self-supervised manner (e.g., BERT), for extracting semantically meaningful word/sentence/document representations in the form of dense numerical vectors. These representations of the text in question are either fed to an appended neural classification head, or are being analyzed by an independent neural network which has been separately trained for classification (e.g., an LSTM, or even a simple MLP). Representative examples include [59], for detecting human trafficking through on-line customer reviews, and [60], which extends this concept to a semi-supervised setting for analyzing hashtags in social media posts, in order to identify on-line drug dealers. An alternative approach in [61] exploits multimodal fusion, by proposing a DNN which combines and analyzes text encodings from BERT and image representations from a pretrained CNN, so as to recognize drug dealers operating in social media.

Besides text classification, Named Entity Recognition (NER) is also very prominent in the fight against illicit goods trafficking. It is a separate NLP task that can be either employed as a preprocessing step, or used on its own. It concerns the automatic identification of novel words as referring to named entities (persons, locations, product brands, etc.). The relevant state-of-the-art relies on neural networks [62], but auxiliary exploitation of external knowledge bases (e.g., Wikipedia) via entity linking is also common [63]. The NER outputs can subsequently be employed for identifying criminal events, modelling criminal groups and discovering the structure of trafficking networks.

A widespread issue in NLP technologies is the scarcity of large datasets for training appropriate DNNs in most languages other than English. Currently, the common solution employed when this problem arises is to utilize multilingual automated translation technologies – also powered by DNNs – as a preprocessing step, in order to convert all inputs to English before analyzing them. Alternatively, cross-lingual NLP can handle multiple languages and transfer knowledge from high-resource languages (like English) to low-resource ones [64].

A3. Network traffic analysis

A dedicated body of research concerns Darknet network traffic analysis. This typically means recognizing Tor network packet exchanges and classifying them with regard to the application/service type. This is a rather difficult endeavor, due to the heavy encryption employed by software such as Tor. The state-of-the-art approaches employ DNNs to achieve this, by analyzing both communication patterns/metadata and the encrypted payload. Examples include [65] and [66], where custom combinations of CNNs and LSTMs are proposed. In this vein of work, the neural (self-)attention mechanism has been indicated as critical for achieving good accuracy. Such an analysis can be combined with more direct visual processing [67] or NLP [68], so that specifically illegal

activities can be identified.

B. Graph analytics

Graph analytics/mining algorithms form the heavy AI machinery when it comes to discovering and identifying trafficking networks that follow digital trends. The relevant graphs mostly fall into three categories:

- **Social network graphs:** These graphs represent relationships between individuals, groups or organizations (nodes) in a social context. They focus on capturing social interactions, friendships, collaborations, hierarchies and other social ties (edges). They can be constructed based on data from on-line social media platforms, known or suspected real-world social relationships between persons, or both.
- **Cryptocurrency transactional graphs:** These are graphical representations of the transactions that occur within a cryptocurrency network, such as Bitcoin, Ethereum, Monero, etc. These graphs capture the flow of funds (edges) between different cryptocurrency addresses/wallets (nodes) and provide insights into the transactional behavior within the network. They can be constructed based on the publicly available transaction data stored in the cryptocurrency's blockchain, i.e., a decentralized, encrypted and transparent data structure that records all transactions within the cryptocurrency network.
- **Knowledge graphs:** These graphs represent structured factual knowledge about entities and their relationships in a specific domain. In general, they can be considered to arise from a hybridization of graph theory with symbolic AI. Nodes express entities (e.g., people, places, events or concepts) and edges logical connections between them. Entities and connections are typically instantiations of a predefined domain-specific ontology, which permits automatic formal reasoning to be applied on the graph once it has been defined. Knowledge graphs can be populated based on inputs from various and diverse sources, such as the outcomes of the DNNs presented in Section III.A, outcomes of social network graph or cryptocurrency transactional graph analysis, other automated processing and/or manually entered information (e.g., by LEA officers).

Several different types of graph analytics/mining algorithms can be applied to these three graph types. The main relevant tasks are briefly reviewed below, organized per graph type.

B1. Social network graphs

The most common task related to combatting illicit goods trafficking is “node classification”. For instance, the method in [69] trains a GNN to classify social media users as drug dealers or not, based on their on-line relationships and the content they post on-line. Similar approaches could be applied to real-world social networks, as well. These can be modelled using gathered intelligence and crime script analysis methods, which aid in identifying criminal *modi operandi*, recognizing

specialized skills and mapping organizational structures [70]. More often, though, real-world criminal social networks are reconstructed semi-automatically by LEAs through phone communication records. Thus, the method in [71] applies GNNs to such a network.

An equally important task is “community detection”, i.e., identifying groups or communities of closely connected nodes within a criminal social network. Uncovering clusters of individuals or entities with strong interactions is a cornerstone of unraveling criminal networks. Classical graph partitioning algorithms based on minimum-cut can be employed, but modularity maximization alternatives, such as the Louvain method [72], are among the most popular choices (e.g., in [71]). Graph-theoretic centrality measures, which quantify the importance of nodes within a graph based on their connectivity, may not only identify nodes that play significant roles in the network, but can also be employed for community detection; this is how the popular Girvan–Newman algorithm operates [73]. Thus, the method in [74] exploits centrality measures within a custom graph clustering algorithm, for identifying communities of criminal actors in social media that engage in human, firearms and drugs trafficking. However, node “influence estimation” is an important task in itself. For instance, centrality measures have been employed for recognizing influential suspicious domains in the Darknet, using graphs where the nodes are Tor hidden services [75].

The task of “link prediction” is particularly significant in criminal social network analysis, since a portion of critical graph connectivity information is typically missing/unknown; even intentionally hidden by traffickers to obstruct LEA investigations. Link prediction algorithms identify missing or future edges and, therefore, can be used to propose potential relationships between suspects or criminal actors, which have not been explicitly observed. For instance, the method in [76] proposes a simple link prediction algorithm, as a preprocessing step for improving community detection. Similarly, the survey in [77] concludes that the most robust link predictor for organized criminal networks is the Katz index, which assigns a score to potential links based on the number of paths that connect them and the lengths of those paths. However, methods based on DNNs tend to become the norm in recent years, such as GNNs [78] or Deep Reinforcement Learning [79].

B2. Cryptocurrency transactional graphs

The transparency of cryptocurrency transactions that rely on blockchain technology allows the construction of an evolving graph, where user addresses form the nodes and fund flows are the edges. Analyzing these graphs, which exhibit small-world properties [80], is extremely important for combatting digitally-aware illicit trafficking groups that operate via the Dark Web. Most tasks that are described in Section III.B1 in relation to social networks are also valid in the case of cryptocurrency transactional graphs as well, with several variations of relevant well-known algorithms having been developed for this scenario. An example would be to use GNNs for classifying nodes, as involved in illicit transactions

or not [81] [82]. Optionally, simple, handcrafted rule systems can then be developed, by exploiting the domain knowledge of LEA experts. These rules sets will provide desired predictions based on the outputs of the machine learning modules. Such a pipeline is depicted in Fig. 4.

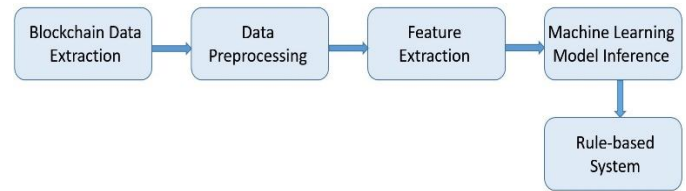


Fig. 4. An example pipeline for automated analysis of cryptocurrency transactional graphs.

However, more specialized methods also exist for tracing cryptocurrency payment flows, i.e., the main interest of LEAs. In criminal investigations, these can be employed to circumvent the anonymity provisions offered by Bitcoin and many later cryptocurrencies. Two such tasks are the most important ones: “address clustering”, which refers to identifying a cluster of different addresses/wallets that can be linked with a single human actor, and “attribution tagging”, i.e., discovering context information and assigning it to an address or transaction. Tags can flag addresses associated with illegal activities, known criminals, or suspicious behavior. Address clustering and attribution tagging operate synergistically. For example, a single tag attributed to a wallet address may lead to its real-world identification (e.g., IP address) and, consequently, effectively de-anonymize an entire cluster of wallets.

Various heuristics have been proposed for address clustering over the years, with the most prominent one being the Multiple-Input Heuristic (MIH): if two addresses serve as inputs in one transaction, and one of these addresses along with an independent third one serve as inputs in a separate transaction, then most likely the three addresses are controlled by a single individual, who possesses the private keys for all of them [83]. However, besides such heuristic rules, typical node clustering algorithms relying on machine learning have also been applied for address clustering [84], while similar approaches have been combined with anomaly detection to automatically flag potentially illicit transactions [85].

Such methods have been successfully employed to trace suspicious monetary flows in cryptocurrency funds. For instance, a criminal network of human traffickers has been identified by correlating specific Bitcoin transactions with sex ads [86]. As a result, novel “mixing services”, which obfuscate fund flows by mixing together irrelevant transactions (e.g., the CoinJoin mechanism), and alternative cryptocurrencies with enhanced privacy support have surfaced in recent years, becoming popular with criminals [87]. State-of-the-art research has been attempting to combat this phenomenon by supplementing graph mining with network analysis methods [88] [89] or with additional features to enhance clustering in the presence of CoinJoin transactions

[90].

B3. Knowledge graphs

A knowledge graph is constructed in two steps: first a domain-specific ontology is defined and, subsequently, the graph is populated based on it, using existing structured knowledge bases (e.g., LEA databases) and the outcomes of entity/relationship extraction from unstructured data. In the case of fighting illicit goods trafficking networks, the ontology typically has to be defined in collaboration with criminologists and LEAs. The methods presented in Sections III.A, III.B1 and III.B2 may serve as automatic mechanisms for extracting entities and relationships from massive volumes of data.

The main benefit of a structured knowledge graph is that it facilitates automatic query answering and formal reasoning, which permit automatic information retrieval and inference of new knowledge. The disparate sources which were previously exploited to populate the graph are implicitly fused in the process, which allows also the automatic detection of conflicts between different sources [91]. Moreover, different knowledge graphs constructed from heterogeneous sources can be merged via algorithms for entity and ontology alignment [91]. Overall, these functionalities can be used for a variety of purposes: automatically linking digital traces (e.g., on-line activities or communication patterns) to physical identities or potential suspects, providing LEA investigators with actionable intelligence (e.g., structure of the trafficking network, key actors, *modus operandi*, or potential vulnerabilities), predictive analytics (e.g., anticipation of criminal behavior, identification of potential threats), criminal profiling and risk assessment of suspects, decision support in investigations, as well as visualization of trafficking network structure (clusters/communities, central actors, hierarchies).

This ambitious vision has already been partially materialized in relevant research, but typically exploiting only a limited number of data sources and modalities. Thus, the method in [92] exploits a custom ontology and a knowledge graph formalism for facilitating criminal investigations and automating the generation of digital evidence admissible in court, while taking care of the chain of custody. However, it focuses exclusively on social media users and exploits only content automatically crawled from such platforms. A partially similar system is presented in [93], where the knowledge graph is populated by automatically crawling and analyzing on-line newspapers, instead of social media content. In a parallel direction, the system in [94] populates the knowledge graph only via semi-automatic Surface Web crawling, while allowing the non-technical user to define/personalize the domain (e.g., illicit firearms trafficking), the ontology, the crawling keywords and the semantic queries. On the other hand, the method in [95] implements a different trade-off: it exploits more varying data sources, such as official crime reports, socioeconomic information and geographic information, which are manually entered into an RDF knowledge graph, instead of automatically crawled, in order to facilitate the discovery of criminal patterns (exclusively for Mexico City). Other

approaches still try to exploit even more heterogeneous data sources, but limit their scope to specific criminal cases [96].

Among the most complete related research prototypes are the ones presented in [97] and [98]. The first one populates the knowledge graph via automatic analysis of the Surface Web, the Dark Web and cryptocurrency transactions, while it has been shown to be able to identify connections between illegal trafficking of different types of goods. The second one is even more heterogeneous: it automatically collects information from the Surface Web, the Dark Web, social networks, financial data, road traffic data, geospatial data, etc.

IV. ETHICAL AND LEGAL ISSUES

AI-powered law enforcement relying on the methods presented in Section III has the potential to create (unintended) harms. Since it is meant to be a public good, it must be regulated and subjected to legal constraints. It is therefore crucial that ethical considerations, legal compliance, and societal acceptability are considered during the design process of similar AI systems, along with potential mitigation measures. Notably, advances in AI itself are gradually making it possible to deal effectively with a subset of these threats. This Section first reviews the relevant ethical and legal issues, before proposing the use of “Trustworthy AI” to overcome a number of them through technical means.

A. Ethical and legal considerations

There are three major areas where ethical and legal gray zones emerge: (a) **Human Dignity and Autonomy**. Human dignity entails that every human being possesses an inviolable intrinsic worth;⁴ respecting this entails all people being treated as individual moral subjects, rather than mere objects.⁵ Representing humans using AI reduces individuals to data points, thus affecting their dignity. Datafying people is a corporeal fetishistic interpretation of the body that abstracts individuals to an “arrangement of static code” [99]. This contributes to a real risk that operators begin to think of people as data points, which is a reductionist activity and can challenge the dignity of both operators and subjects of AI analysis.

Human dignity and autonomy could also be threatened by graph node classification, particularly from social network graphs and knowledge graphs. Node classification focuses on capturing social interactions to represent social relationships. Data produced through online activities, particularly social networks, are used to construct online “associative networks” [99]. Constructing digital “relationships” challenges the inherent complexity of human interactions, while simultaneously refusing to see an individual’s autonomous associations. Associating to a community is often vital to constructing and expressing one’s humanity and identity [100]

⁴ European Union, *Charter of Fundamental Rights of the European Union*, 26 October 2012, 2012/C 326/02, Article 1. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:12012P/TXT>.

⁵ European Commission, *Ethics By Design and Ethics of Use Approaches for Artificial Intelligence*, 2021. Available: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ethics-by-design-and-ethics-of-use-approaches-for-artificial-intelligence_he_en.pdf [hereinafter EC, Ethics by Design].

[101]. Instead, results of node classification, and community detection methodologies, can be viewed as representative and actionable, rather than the human-created autonomous reality.

(b) **Bias and Discrimination.** In an AI context, equality demands that system’s operations cannot generate unfairly biased outputs.⁶ AI systems should, ideally, not include historic bias. The continuation of such biases could lead to unintended (in)direct prejudice and discrimination.⁷ Ensuring training and operational data is accurate, representative, reliable, relevant, and up-to-date with the population that the AI system will be applied to can be helpful in mitigating biases.⁸ Yet, considering the massive, unsupervised gathering and analysis of both unverified and subjective data from the Surface Web, Dark Web, and social media sites for DNNs and graph analytics, the potential for biased data to be included remains rife.

Analysis of social networks and knowledge graphs, particularly the Katz index, can result in “social sorting”. This involves sorting individuals into categories, and assigning worth or risk to individuals [102]. Such profiling can constitute a serious data protection issue if the relevant safeguards are not followed.⁹ This practice can also threaten fundamental rights such as dignity, equality, and integrity by classifying some individuals above or below others [102]. It is especially problematic where classifications are based on incorrect or biased information, resulting in discrimination [103]. This can then threaten the freedom of association and expression, and even incite behavioural change in the individuals’ online actions, so as to avoid being profiled and sorted by such methodologies.

Predictive AI methods, like DNNs, can contribute to a feedback loop of biased outcomes. If DNNs are trained on biased data, there is a risk that historic biases continue or worsen with the technology deployment. This could have normative impacts, including, continuation and codification of discriminatory practices towards certain individuals, which could potentially exacerbate prejudice and marginalization in society.¹⁰

(c) **Privacy and Data Protection.** The processing of crawled open-source data to train DNNs has the potential to engage both privacy rights and data protection regulations, due to the vast amounts of publicly available personal data that can be gathered. Such data is still subject to data protection regulation and must be processed lawfully. Where information

related to an identifiable person is subject to any operation, personal data is being processed;¹¹ this might include, for example, collecting details of cryptocurrency addresses/wallets that can be linked to their owners (even if this is very difficult to do).

A key legal issue concerns identifying an appropriate legal basis for collecting and processing open-source personal data. For researchers crawling personal data, the complexities of gathering consent at scale make that legal basis impractical. However, the public task legal basis¹² might be relevant where national law allows this. The legitimate interest legal basis could also be relevant where the processing is for scientific research, and does not override “the interests or fundamental rights and freedoms of the data subject”;¹³ though compliance requires a legitimate interest assessment that very much depends on the reasonable expectations of the persons concerned.

Social media users might not reasonably expect their data to be collected and used to train DNNs. Private online spaces should be considered private insofar as a user has a reasonable expectation that they control who sees the information they share within that space; this is especially the case where site rules give the perception that users can tightly control access to their data.

B. *Trustworthy and Technically Robust AI for Ethical Countermeasures*

The widespread deployment of the methods presented in Section III has the potential to bring about various detrimental side-effects for both citizens and organization, such as the ones described in Section IV.A. In order to proactively mitigate similar dangers, several initiatives have been set up to establish the principles of a trustworthy and secure AI. For example, the European Commission (EC) has created an independent expert group, namely the High-Level Expert Group on Artificial Intelligence, which prepared the “Ethics Guidelines for Trustworthy Artificial Intelligence (AI)” document¹⁴. It is a framework that aims to secure fundamental citizen rights, formulate ethical principles and make current AI systems trustworthy from societal, ethical, and legal perspectives. The same year, ACM released the ACM Code of Ethics and Professional Conduct¹⁵, covering ethics across the computing field: the tech code of ethics, computing ethics, software ethics, programming ethics, AI ethics, etc. Prior to this, IEEE launched the IEEE Global Initiative for Ethical Considerations in Artificial Intelligence and Autonomous Systems with the mandate to provide a practical guide¹⁶ for addressing the urgent relevant ethical concerns. All these initiatives paved the way for many ethical frameworks [104] regarding trustworthy AI in various domains, such as digital

⁶ European Commission, High-Level Expert Group on Artificial Intelligence, “Requirements of Trustworthy AI”, 2019. Available: <https://ec.europa.eu/futurium/en/ai-alliance-consultation/guidelines/1#Human%20agency>.

⁷ AI HLEG, The Assessment List for Trustworthy Artificial Intelligence (ALTAI) for self-assessment, 2020. Available: <https://digital-strategy.ec.europa.eu/en/library/assessment-list-trustworthy-artificial-intelligence-altai-self-assessment> (hereinafter ALTAI).

⁸ Information Commissioners Office, How to use AI and personal data appropriately and lawfully. Available: <https://ico.org.uk/media/for-organisations/documents/4022261/how-to-use-ai-and-personal-data.pdf>.

⁹ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC, Article 22. Available: <https://eur-lex.europa.eu/legalcontent/EN/TXT/PDF/?uri=CELEX:32016R0679&from=EN> [hereinafter GDPR].

¹⁰ ALTAI, *supra* note 7 at 16.

¹¹ GDPR, *supra* note 10 at Article 4(1) and 4(2).

¹² *Ibid.* at Article 6(1)(e).

¹³ *Ibid.* at Article 6(1)(f).

¹⁴ European Commission, Directorate-General for Communications Networks, Content and Technology, Ethics guidelines for trustworthy AI, Publications Office, 2019. Available: <https://data.europa.eu/doi/10.2759/346720>.

¹⁵ ACM Code of Ethics and Professional Conduct, 2018.

¹⁶ IEEE Ethically Aligned Design, 2019. Available: <https://ethicsinaction.ieee.org/>.

healthcare [105], telecommunications (6G) [106], or fintech [107], as well as for many types of sectors, e.g., public, private and non-governmental organizations [108].

In order to provide technically robust and trustworthy AI systems, potential solutions should focus on three key aspects: (a) **Transparency of AI models** through distinct interpretability and explainability features, which should be employed for both the design of the AI models and the description of the data used during model conceptualization. Transparency allows software developers to analyze their system, enables security practitioners to trust it and facilitates regulators to ensure it is safe and fair. Unfortunately, in many cases, these AI models become so complex that it is often challenging to understand the reasons behind their predictions/decisions. In general, “black-box” models yield excellent performance in terms of accuracy but users, their designers, cannot fully understand how the underlying variables are combined to reach a decision. This fact brings to the fore the risk of not respecting the fundamental rights of citizens to privacy preservation, to equality before the law, etc. Thus, algorithms have been developed for interpreting and clarifying the decision-making process of AI models, creating the new subfield of explainable AI (XAI) [109]. In this context, new regulations have sprung up requiring human-readable reports of how AI systems deployed in the real world have arrived at their conclusions. One notable example is Europe’s General Data Protection Regulation (GDPR), which stipulates that organizations must disclose “meaningful information about the logic involved” when using personal data for “automated decision-making, including profiling”. All these laws have been motivated by the constantly growing concern that “black-box” systems may be hiding evidence of illegal, or perhaps just unsavory, discriminatory practices [110]. Thus, it is evident how making XAI deployment mandatory by law is pertinent for fostering socially responsible LEA practices.

In order to cope with the aforementioned transparency problems and to comply with EU regulations, engineers are encouraged to follow an explainability-by-design approach during the system’s architectural design [111]. Moreover, development and use of inherently explainable “glass-box” AI models (e.g., models derived from causal analysis, or relying on graph reasoning) should also be encouraged over “black-box” models (e.g., DNNs), in cases where the former ones yield acceptable performance [112]. In cases where “black-box” models are deemed necessary (e.g., due to their high accuracy when large data volumes are available), their explainability must be targeted via well-established XAI algorithms, able to interpret model predictions and boost trustworthiness. Examples of such algorithms include Deep Learning Important Features – DeepLift [113], Local Interpretable Model-agnostic Explanations – LIME [114], Shapley Additive Explanations – SHAP [115] and Causal Explanations – CXPlain [116].

XAI can be leveraged to explain to security practitioners the reasoning behind a predicted mapping between illicit goods trafficking activities to specific criminal networks and, thus, guarantee lawful prosecution through court-proof evidence collected by LEAs. The overall ambition of this process is to remove concerns about false matches, low

fairness and reliability caused by bias mitigation in the utilized AI algorithms. A main concern is to exploit AI transparency for identifying and avoiding the perpetuation of systemic social inequalities, such as racial and gender discrimination, marginalization, etc., which may be reproduced by AI model predictions due to the data they have been trained on. This functionality can be utilized even during the initial construction of AI models, to ensure that the training data are diverse, representative, and free from bias against minorities. Notably, implicitly fusing multiple different information sources on a Knowledge Graph (as discussed in Section III.B3) has the potential to automatically mitigate such biases that are present only in a subset of the utilized input modalities. However, explicitly enforcing AI transparency at the previous stages of the analysis significantly enhances overall trustworthiness and facilitates better compliance with regulations and laws.

XAI can be offered as a library of methods aiming to explain how “glass-box” recommendations are made, or even “black-box” ones up to a degree. However, in complex problems such as the accurate identification of illicit goods trafficking networks, the trade-offs between accuracy and interpretability must be carefully balanced: explaining “black-box” AI model predictions is particularly challenging and non-trivial from a technical standpoint.

(b) **Reliability of AI models**, identifying in early stage any potential vulnerabilities and implementing appropriate technical solutions to guarantee that the overall system will not fail or be manipulated by an adversary [117]. In general, poor real-world accuracy of AI systems and the identification of vulnerabilities leading to malfunctions are the key indicators of AI model unreliability. Overcoming this kind of issues is a prerequisite for AI trustworthiness, but it is not trivial.

In machine learning systems, such as DNNs, overfitting is a major challenge to reliability during real-world deployment, due to low generalization ability in novel data inputs. The issue arises when the AI model has memorized in detail the noise or the random error components within its training dataset, instead of learning the actually desired patterns. Dropout methods [118], data augmentation [119] and batch normalization algorithms can pre-emptively mitigate such problems during the training phase. Dropout essentially turns the trained DNN into an ensemble of multiple subnetworks with different topologies, thus combatting overfitting by averaging their predictions. Data augmentation simply involves artificially increasing the size of the training dataset, while batch normalization standardizes the input of each DNN layer during training to zero mean and unit variance, resulting in a regularizing effect due to the insertion of noise. All these approaches are applied after carefully reducing the complexity of the AI model by eliminating layers or neurons, to minimize susceptibility to overfitting. A complementary aspect of DNN reliability during deployment, also related to maintaining high generalization ability in novel data inputs, is achieving robustness to noisy inputs. A particular, “extreme” instance of this problem is robustness against adversarial attacks, i.e., a particular scenario where a pretrained DNN is fed special inputs with minimal noise, carefully and explicitly added so that the AI model is fooled. Currently, properly adapting the

DNN training objectives seems to be the best pre-emptive strategy for ensuring deployment-time robustness [120].

Another aspect of AI reliability is the elimination of any potential bias in the algorithms, e.g., detected using XAI approaches. Notably, the terms bias, discrimination and unfairness are often used interchangeably with similar meanings. AI bias is becoming more apparent and problematic with the wider use of AI-based decision support systems. One of its negative consequences is discrimination, which refers to the unfair or unequal treatment of individuals based on certain characteristics [121]. Special algorithmic mechanisms can be utilized to ensure the fairness of AI models, with the simplest ones being systematic methods to curate the training datasets for avoiding biases that may unfairly target specific demographics or regions. Technical schemes such as data augmentation and data balancing can be exploited to ensure fair representation from different categories of illicit goods and trafficking scenarios. The relevant literature often links discrimination with bias, which refers to a deviation from the standard that is necessary for identifying statistical patterns in the data. Simultaneously, reducing bias and discrimination is directly linked to the perception of justice [122]. Understanding AI decision-making processes becomes paramount for justifying decisions, especially in safety and security cases. However, these processes often do not provide information relevant to judgments of justice, which can compromise the fundamental accountability of human decision-making.

In the context of the methods presented in Section III, AI can be used in courts to assess whether a defendant has committed or will recommit a crime, in criminal identification through facial recognition from CCTV, etc. [123]. Given that such applications of AI have direct consequences in citizens' lives and can become harmful if designed without taking into considerations the fairness principle [124], [125], the value of AI reliability becomes evident. Yet, several studies have shown that satisfying multiple notions of fairness simultaneously is an impossible task; individual and group fairness may occasionally be incompatible [126] [127].

(c) **Protection of data in AI models**, through the application of appropriate control measures in order to ensure the citizens' right to privacy. It is essential to address the possibility of deployed AI models having unintentionally memorized personal details unrelated to their primary goal, which could lead to unintended disclosure [128]. This can be problematic, especially when the AI models have been trained with private or sensitive datasets. Sophisticated methods that guarantee data privacy have emerged in the context of trustworthy AI and can be readily applied. For instance, in the EU, anonymization or pseudonymization methods are used to safeguard GDPR compliance of AI systems. Advanced AI algorithms, e.g., for face de-identification in image datasets [129], can be exploited for enhancing privacy guarantees, while differential privacy methods may be deployed to prevent re-identification of individuals through the AI system's outputs [130]. Moreover, privacy-by-design AI methods such as federated DNN learning can be employed to protect sensitive data [131], while allowing collaboration between different LEAs. Finally, besides technical solutions, organizations can complete a "Data Protection Impact Assessment" (DPIA) to

evaluate privacy and data protection impacts of AI systems, even during the system design phase. The goal is to identify and address potential issues associated with high-risk instances of data processing; in this case, with a particular emphasis on applications related to public security and use by LEAs.

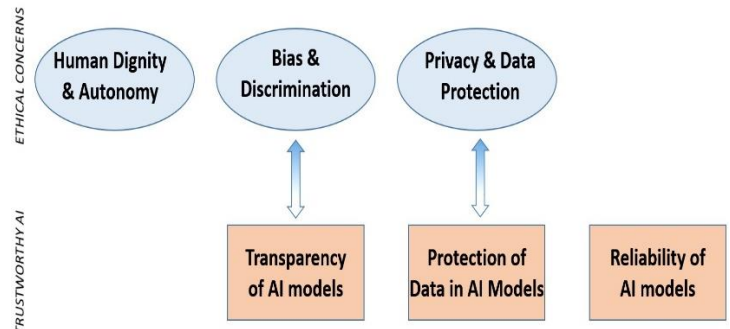


Fig. 5. A mapping of ethical/legal concerns and technical Trustworthy AI means, regarding the large-scale deployment of AI by LEAs trying to combat illicit goods trafficking.

The overall situation is graphically summarized in Fig. 5. As it can be seen, technical Trustworthy AI solutions cannot be deployed to properly handle human dignity and autonomy issues; a suitable legal, cultural and operational framework is required to achieve this. On the other hand, Trustworthy AI can be employed to enhance the reliability of AI models that are deployed in the field by LEAs; this does not respond to any ethical or legal concerns, but is nevertheless a prerequisite for large-scale social and operational acceptance of AI-enabled solutions. Finally, technical means for transparent AI and for data protection in AI are able to directly address relevant ethical concerns.

V. CONCLUSION

Overall, recent hi-tech trends employed by the organized crime for illicit goods trafficking can be efficiently combatted, using dedicated versions of state-of-the-art AI methods for large-scale information processing. This is important due to the increasing digital sophistication of trafficking networks and the damage induced by their practices (social, economic, or even environmental).

Proper deployment of such AI tools by LEAs constitutes a practical way to deal with the emerging criminal modus operandi that have been identified in this paper. Although various legal and ethical considerations unavoidably arise, a subset of these issues can be mitigated by technical means, such as the Trustworthy AI paradigm. Evidently, said means should be properly integrated into a wider, efficient matrix of regulations and procedures, devised to guarantee ethical operation of AI-powered law enforcement.

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